



Nebraska Public Power District

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10 CFR 50.73

NLS2024038
May 13, 2024

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2024-005-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2024-005-00.

This letter does not contain regulatory commitments.

Sincerely,



Khalil Dia
Site Vice President

/jo

Attachment: Licensee Event Report 2024-005-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via IRIS entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104EXPIRES: 04/30/2027 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.						
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)											
1. Facility Name Cooper Nuclear Station					☒ 050 ☐ 052		2. Docket Number 298		3. Page 1 of 4		
4. Title High Pressure Coolant Injection Pressure Switch Installation Causes Condition that Could Have Prevented Fulfillment of a Safety Function and a Condition Prohibited by Technical Specifications											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050 ☐ 052	Docket Number
03	13	2024	2024	005	00	05	13	2024	Facility Name	☐ 052	Docket Number
9. Operating Mode 1						10. Power Level 100					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ OTHER (Specify here, in abstract, or NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Linda Dewhirst, Regulatory Affairs and Compliance Manager								Phone Number (Include area code) (402) 825-5416			
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable to IRIS		Cause	System	Component	Manufacturer	Reportable to IRIS	
B	BJ	PS	S345	Y							
14. Supplemental Report Expected						15. Expected Submission Date			Month	Day	Year
☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)											
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines) On March 13, 2024, while performing planned maintenance on High Pressure Coolant Injection Pressure Switch 2787 (HPCI-PS-2787), the pressure switch setpoint was found to be outside of calibration tolerance. A new pressure switch, that passed calibration tolerance during bench testing, was subsequently installed and found to be out of tolerance. Further investigation revealed that the pressure switch had not been installed per the seismic test report. As such, the pressure switch was removed and reinstalled per the seismic test report and passed calibration testing. Guidance documents directing installation of the pressure switch will be revised to include installation requirements. Event Notification 57041 was submitted on March 21, 2024, for this condition. There was no impact on nuclear safety, plant reliability, radiological safety or industrial safety.											



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Cooper Nuclear Station	2. DOCKET NUMBER 05000-298	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
		2024	005	00

NARRATIVE

PLANT STATUS

At the time of the event on March 13, 2024, Cooper Nuclear Station (CNS) was in Mode 1, Power Operation, at 100 percent steady-state power.

BACKGROUND

The High Pressure Coolant Injection (HPCI) system [EIS:BJ] provides protection to the core in case of a small break in the reactor coolant pressure boundary which does not result in a rapid depressurization of the reactor vessel [EIS:RPV]. The HPCI system permits the nuclear plant to be shutdown while maintaining sufficient reactor vessel water inventory until the reactor vessel is depressurized. HPCI continues to operate until reactor vessel pressure is below the pressure at which Low Pressure Coolant Injection system [EIS:BO] operation or Core Spray system [EIS:BM] operation can be used to maintain core cooling.

HPCI consists of a steam turbine assembly [EIS:TRB] driving a multi-stage booster and main pump assembly [EIS:P] and system piping, valves, controls and instrumentation. The HPCI turbine is driven by steam from the reactor which is generated by decay and residual heat.

The function of the HPCI Pressure Switch (PS) is to operate the auxiliary oil pump. It senses oil pressure at the turbine driven oil pump discharge. HPCI-PS-2787 opens when the turbine driven oil pump pressure is greater than or equal to 85 psig, stopping the auxiliary oil pump. HPCI-PS-2787 closes when the turbine driven oil pump pressure is greater than or equal to 20 psig, thus starting the auxiliary oil pump at a shutdown speed.

HPCI-PS-2787 was installed as an Equipment Qualification pressure switch as part of original station equipment. On March 26, 2009, HPCI-PS-2787 was replaced with the same model of pressure switch. This replacement was the last remaining spare Model 9012-ACW-22. This model switch is obsolete and no longer available.

In 2012, the vendor manual for the switch was updated and included installation torque values for the new 9012G model.

In July 2014, an equivalent change (EC) was approved to replace HPCI-PS-2787, Model 9012-ACW-22, with a new Model 9012G. The EC did not include discussion on mounting requirements for this new model. A test report was issued for Model 9012G on August 30, 2016, and included mounting requirements for Model 9012G.

In 2023, HPCI-PS-2787, Model 9012-ACW-22, was replaced with Model 9012G. The work documents for this replacement did not specify a torque value for the mounting screws.



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HPCI is a single train system and is required to be operable to satisfy the requirements of Technical Specifications (TS) 3.5.1.

EVENT DESCRIPTION

On March 11, 2024, at 22:29 hours, Operations personnel declared HPCI inoperable for upcoming planned maintenance. On March 13, 2024, during the planned HPCI system work window, the setpoint for HPCI-PS-2787 was found to be out of calibration tolerance at 47.12 psi. The calibration tolerance is required to be greater than or equal to 85 psi.

A replacement pressure switch was consequently installed on March 13, 2024. The new pressure switch was bench calibrated to trip at 95 psi and then installed. After installation, the switch was tested and the setpoint was found to be 44.7 psi, and when reset, 32.4 psi, thus not meeting the calibration tolerance. Upon investigation, it was discovered that the tightness of the pressure switch mounting screws drastically affects the calibration of the switch. Further research determined that the pressure switch had not been installed per the seismic test report mounting requirements when the new model 9012G was installed in 2023. The rubber vibration pad, located between the switch body and the mounting plate, had not been removed prior to installation in 2023 and 2024. Documentation shows that the rubber vibration pad and the mounting hardware of the original was reused for installation since there was not a new pad or hardware included with the replacement switch.

After the March 13, 2024 calibration, discussion with the vendor confirmed that the installation torque values for the model 9012G pressure switch should not be exceeded and a vibration pad should not be installed with the switch. Exceeding mounting torque requirements or installing a vibration pad can cause setpoint changes after calibration. As such, the rubber vibration pad was removed and the proper torque was applied to the mounting hardware restoring the setpoint to the proper value. Operations declared HPCI operable on March 15, 2024, exiting LCO 3.5.1, Conditions C and D.

The pressure switch and rubber pad were sent to a vendor for testing to determine if the effect could be reproduced. Testing reproduced the setpoint changes if the mounting hardware was over torqued with the rubber pad installed. Removing the rubber pad and applying the proper torque to the mounting hardware restored the setpoint to the proper value.

BASIS FOR REPORT

The event is reportable under 10 CFR 50.73(a)(2)(i)(B) as any operation or condition prohibited by Technical Specifications as well as 10 CFR 50.73(a)(2)(v)(D) as any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

Event Notification 57041 was submitted on March 21, 2024, for this condition.



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SAFETY SIGNIFICANCE

There were no actual consequences to general safety of the public, nuclear safety, industrial safety, and radiological safety for this event.

This condition involved a single train safety system that could have prevented the fulfillment of a safety function needed to mitigate the consequences of an accident. However, the condition has been previously analyzed and does not represent a new failure mode.

A subsequent engineering evaluation demonstrated that the condition would not inhibit the ability of HPCI-PS-2787 to perform its safety function. Accordingly, this event will not count toward the NRC Safety System Functional Failure performance indicator.

CAUSE

The cause of the event was the installation and mounting of the new model pressure switch was not in accordance with the vendor mounting recommendations and the seismic test report, resulting in a shift in calibration which changed the setpoint outside of the acceptable range.

CORRECTIVE ACTIONS

Guidance documents will be revised to include the installation requirements for the pressure switch as well as seismic testing configuration.

PREVIOUS EVENTS

A review of CNS licensee event reports (LER) for the last three years was performed. There were no previous occurrences involving a similar failure to that discussed in this report. The following conditions related to HPCI have been reported.

LER 2024-003-00 – HPCI declared inoperable due to drain line piping steam leak [April 22, 2024].

LER 2022-003-00 – Inadvertent HPCI injection into the RPV during HPCI beginning of cycle surveillance testing. The injected water increased reactivity and caused a reactor scram [January 11, 2023].

LER 2021-003-00 – During surveillance testing, HPCI flow indicating switch, HPCI-FIS-78, failed to trip prior to exceeding the TS 3.3.5.1 (Emergency Core Cooling System Instrumentation) function for HPCI pump discharge flow-low (bypass) [December 27, 2021].